

DECODED EXPLAINED

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Pyroprocessing: heat shows the way

Vasudevan Mukunth

Pyroprocessing is a way to change a solid material physically or chemically using high temperature. It is a dry process and very energy-intensive. The three sectors that use it most are cement-making, metallurgy, and nuclear power.

Of these, cement-making uses pyroprocessing the most. First, finely ground limestone, clay and iron — the raw materials — are fed into a rotary kiln. As the temperature increases to 900°C , the limestone loses its carbon dioxide, and at around $1,450^{\circ}\text{C}$, the mix partly melts to form marble-sized nodules called clinker. The clinker is then ground to produce cement.

In metallurgy, pyroprocessing is used to extract metals from their ores in multiple stages. For example, sulphide ores are heated in air — or roasted — to convert them into metal oxides, like zinc sulphide. Smelting melts an ore to separate the metal from waste impurities called slag. Heating limestone to yield lime is likewise called calcining.

In the nuclear industry, pyroprocessing has a slightly different meaning. It is used to reprocess spent nuclear fuel, in techniques developed in the 1980s and 1990s. First, used nuclear fuel is broken up into pieces and placed in a salt bath — usually a mixture of lithium and potassium chlorides at 500°C or more. Then, an electric current is passed through the salt bath, causing



A rotary kiln is visible as the rust-coloured horizontal tube at the centre in this image of a cement plant in the U.S. GREG GOEBEL (CC BY-SA)

different elements to become separated depending on their electrochemical properties. Operators then recover the elements of interest in separate streams.

This form of pyroprocessing has been the subject of study in Japan, South Korea, and the U.S. and is used as part of programmes involving advanced fast reactors.



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1. Consider the following statements regarding pyroclastic materials:

1. They are fragmented materials ejected during volcanic eruption
2. Volcanic ash particles are larger than volcanic bombs.
3. Tuff is formed by the consolidation of volcanic ash.

Which of the statements given above is/are correct?

- (a) 1 and 2 only
- (b) 1 and 3 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

Answer:

- (b) 1 and 3 only

Rock marks, game boards: Wildlife trail leads to prehistoric past in Aravalli

**Sophiya Mathew
& Abhimanyu Hazarika**
New Delhi, June 7

FOUR MONTHS ago, a wildlife trail through the rocky outcrops of the Bhondsi stretch of the Aravalli forests in Gurgaon led

the wildlife enthusiasts to a stone surface marked with dozens of carefully carved cup-shaped depressions, geometric grids, and a large foot-shaped engraving.

That spot has since been identified as a cluster of ancient petroglyphs (images or designs

carved or engraved on rock), possible stone game boards, and stone tools — part of a wider archaeological pattern emerging from the Aravalli belt where, researchers say, evidence from multiple prehistoric phases appears to survive within the same landscape.

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"A chronological continuity is being found here," said

Akash Gupta, a PhD scholar and faculty member at Delhi University's Satyawati College who has examined the Bhondsi material. "The discovery is significant because it

shows a rare continuity of human activity linking early tool-making traditions with later ritual, symbolic, and social life across this region."

Gupta said scholars such as French archaeologist André Leroi-Gourhan viewed hand engravings on rock as evidence of a cognitive shift. Placing the latest discovery in context, he said the region has yielded signs ranging from the Lower Palaeolithic to the Mesolithic period.

"Handaxes and cleavers point to early stone-tool traditions, while the tools indicate later Middle Palaeolithic activ-



(Left) Foot engraving on quartzite rock; an Acheulean bifacial hand axe found close to rock art cliff. COURTESY: TEJVEER MAVI, SHAILESH BAISLA AND YATIN VERMA

ity (300,000 to 50,000 years ago). The cupules (cup-shaped depressions) and rock-art-like markings are more likely to belong to later symbolic or ritual phases, especially the Mesolithic, when prehistoric communities are known to have left behind paintings, burial practices and other markers of social life," he said.

Officials said excavation teams from the Chandigarh Circle of the Archaeological Survey of India (ASI) and the Haryana Archaeology Department have been documenting the site, including the petro-

glyphs and other remains, and have submitted a report.

Kamei Athoilu Kabui, Superintending Archaeologist with ASI's Chandigarh Circle, said, "The site needs to be protected from the land mafia and further steps in this regard need to be taken by the state government department."

She said the ASI's local exploration team, working with local villagers, had found in situ petroglyphs and stone tools.

Anil Tiwari, Deputy Superintending Archaeologist, ASI Chandigarh, said: "A brief report was prepared but there is

still a lot of scope for further work and research."

According to Shalish Baisla, a PhD scholar and researcher at the School of Heritage Research and Management, Dr B R Ambedkar University Delhi, the Bhondsi documentation began after wildlife photographers Yatin Verma and Ramkumar shared images of the cup-marked rocks they had come across in the forest in February.

Verma said, "During a bird-watching trip with my team, we heard a leopard calling from a cliff above us. The cliff is a

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Which one of the following prehistoric sites is associated with the earliest evidence of agriculture in the Indian subcontinent?

- (a) Bhimbetka Rock Shelters
- (b) Mehrgarh
- (c) Hunsgi
- (d) Bagor

Answer:

(b) Mehrgarh

From borderland to India's strategic resource frontier

Within days of one another, official platforms of the Ministry of Mines cast several northeastern States in a similar frame – as repositories of strategic minerals and untapped potential. Manipur was described as a “quiet mineral frontier”, Arunachal Pradesh as a “resource-rich frontier”, while Meghalaya and Mizoram were portrayed through comparable narratives that emphasised the hidden wealth beneath their hills. Governments routinely publicise natural resources and development opportunities, and such descriptions would ordinarily attract little attention.

Taken together, however, they point to a broader shift in the language through which the northeast is increasingly being framed in the national conversation and strategic picture.

The critical mineral push

The timing is significant because critical minerals have moved from geological discussions into strategic ones. Lithium, cobalt, graphite, nickel and rare earth elements increasingly shape industrial competition, technological manufacturing and energy transitions. Batteries, semiconductors, renewable technologies and defence systems depend upon them, and countries have begun repositioning themselves around access to these resources. India itself continues to depend on imports for several critical minerals and has consequently expanded exploration efforts. According to a Ministry of Mines reply in Parliament, the Geological Survey of India undertook 43 critical mineral exploration projects across northeastern States during the 2022-23, 2023-24 and 2024-25 field seasons, covering minerals such as graphite, vanadium, lithium, rare earth elements, nickel and cobalt. Exploration activity has expanded across Arunachal Pradesh, Meghalaya, Assam, Nagaland and Manipur. In Manipur, projects involving nickel, cobalt and chromium exploration have recently been initiated.

Geological surveys have pointed toward mineral potential across the region for years.



Sangmuan Hangsing

Researcher and alumnus of the Kautilya School of Public Policy

Critical mineral ambitions must account for the people, land and history of northeast India

What appears to be changing is the language through which that significance is increasingly being understood. The northeast has long held strategic significance that extends beyond geology, but the framework through which that significance is understood now appears to be widening.

Shift in language

For decades, the northeast has figured in national strategy largely through the language of borders and security. Discussions have centred on insurgencies, territorial management, connectivity initiatives and geopolitics considerations involving neighbouring countries, while infrastructure and development were often justified as instruments of strategic access and territorial security.

The language of resources is increasingly entering a strategic space once dominated by concerns over borders and security. Critical minerals are now discussed alongside trade corridors and geopolitical access, with territorial and resource security converging. Places once viewed mainly as sensitive border regions are increasingly seen as strategic assets.

The repeated use of the word frontier is revealing, because frontiers rarely function as neutral descriptions. They do not merely describe geography; they often reflect how States imagine it. Historically, frontiers have been viewed as spaces awaiting integration, development or extraction because they appear as landscapes of future possibility.

The difficulty is that frontiers are rarely empty spaces waiting to be discovered. The hills and valleys of the northeast already contain dense social and political worlds structured around customary land systems, local institutions and long-standing relationships with territory. Questions of land often extend beyond economics, as they are also tied to authority, identity and memory. Resource extraction thus enters landscapes that already possess institutions and histories of their own.

These questions become particularly significant in regions where political uncertainties continue to shape everyday life. In Manipur, years of violence and displacement have intensified debates over land and territorial arrangements. Similar concerns about ownership, ecological vulnerability and local participation have surfaced across the northeast at different times. Projects involving land often acquire meanings that extend beyond development, as communities interpret them through the lens of trust, representation and political inclusion.

Resources and inclusion

India's search for critical resources is understandable within a global environment increasingly shaped by supply-chain uncertainty and strategic competition. The northeast itself also requires infrastructure, employment and economic opportunities that have remained uneven for decades. Questions surrounding resource development rarely fit neatly into positions of support or opposition.

How quickly these transitions unfold and who shapes them may matter as much as the resources themselves. For a very long time, national priorities and local realities in the northeast often moved at different speeds. Connectivity projects sometimes arrived without corresponding economic ecosystems, while strategic considerations frequently overshadowed questions surrounding participation and representation. Resource development risks reproducing similar tensions if extraction begins moving faster than institutions capable of managing its social consequences.

What is being debated extends beyond the minerals beneath the hills. The northeast has spent years being viewed first as a border to be secured and then as a corridor to be connected. If it now begins entering national imagination as a landscape of strategic resources, the question is whether this new frontier will finally include the people who already inhabit it, or merely assign another purpose to the land beneath their feet.

Not seeking mediation on border dispute: Nepal FM

Kathmandu's focus is on establishing claim on tri-junction; new 'political reality' in Nepal under PM Shah not in favour of tying relations with India to 'old baggage', says visiting Foreign Minister Khanal

Kallol Bhattacharjee
NEW DELHI

Kathmandu is focused on establishing Nepal's claim on the Kalapani-Lipulekh-Limpiyadhura tri-junction and is not "asking for mediation" by third parties, Nepal's Foreign Minister Shisir Khanal said here on Sunday.

At a press conference at the Embassy of Nepal, Mr. Khanal said he represented a "completely new political reality in Nepal" that was determined to deliver "uncompromising good governance". The new government of Prime Minister Balendra Shah was not in favour of tying Nepal-India relations down to "old baggage", he added.

"We want to solve our disputes through diplo-



Field trip: Nepal's Foreign Minister Shisir Khanal visits the Embassy of Nepal in New Delhi on Sunday. ANI

matic processes. We just want to see if we can access some of the documents that might be in libraries or museums in the U.K. Our position was not that we were asking for mediation," Mr. Khanal said, elaborating on Mr. Shah's remarks that Nepal was in contact with China and the

U.K on the Kalapani-Lipulekh-Limpiyadhura dispute with India.

Shah's statement

"The problems that existed when British India left the region still persist, so we believe Britain also has a role to play in this matter," Mr. Shah had said in Ne-

pal's Parliament, stirring debates.

India's Ministry of External Affairs had issued a rebuttal, reiterating that India and Nepal had established a bilateral mechanism for discussing the dispute, and there was no scope for third-party intervention.

As a new-generation political entity, the Rastriya Swatantra Party refuses to "look at India through the distorted, hypersensitive lens of 20th century geopolitics", Mr. Khanal said. "Instead, the RSP wants to shift the entire vocabulary of Nepal-India relations away from geopolitical friction and square it firmly on development diplomacy," he added.

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Poor facilities, stigma force Odisha girls to skip school during menstruation: study

Bindu Shajan Perappadan
BHUBANESWAR

Absenteeism from school during menstruation continues to be a major concern for female students across Odisha, with nearly 74% of them missing anywhere between one to eight days of classes per cycle, according to a study on menstrual health released here recently.

Pain and discomfort were identified as the primary reasons for absenteeism from school. BISWARANJAN ROUT



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urban areas, and was conducted by organisations, including UNICEF, Aaina, WaterAid, AIIMS-Bhubaneswar, and IIT-Bhubaneswar.

While 94% of the schools surveyed had separate toilets for girls, the lack of basic menstrual hygiene support systems, and non-availability of water and soap were cited as major hurdles. The study found that around 56% of schools either lacked menstrual waste disposal facilities

or relied on unsafe disposal methods.

The lack of healthcare staff within schools is another concern, with only 27% of schools having a nurse or health worker and 44% reported having first aid kits.

Over the past three years, public spaces in Bhubaneswar, have been audited by adolescents using the Public Spaces Assessment Framework developed by the World Resources Institute India, a

global non-profit, with inputs from over 800 young people.

“Here, too, one of the key issues that persists is the lack of inclusive and period-friendly toilets in public spaces. Building on three years of field insights, the Safe, Vibrant, and Healthy Public Spaces project offers a model for inclusive toilets that we aim to demonstrate at three strategic locations across Bhubaneswar. The initiatives will be led by adolescent champions of Aaina, seeking to support city agencies to build awareness around accessible and inclusive sanitation infrastructure, especially for women, girls, and persons with disabilities, while ensuring dignity, hygiene and comfort in public spaces for everyone,” Mukti Swaroop Pradhan, programme associate, Urban Development, Sustainable Cities and Transport, WRI India, said.

How ICMR is rewiring the health ecosystem

As the nation advances toward the vision of **Viksit Bharat 2047**, the question is not only about how to treat disease, but how one builds a health system that is anticipatory, equitable, and innovation-driven. At the heart of this transformation lies a reimagined approach to health, one that connects data to decisions and decisions to impact. Building on the hard-earned lessons of the COVID-19 pandemic, the Indian Council of Medical Research (ICMR), India's apex body for biomedical research, has undertaken a series of forward-looking reforms. These reforms range from reimagining its institutional architecture to strengthening the way research is funded and translated into impact. This shift reflects a deliberate move towards integrated, institute-led research aligned with national priorities, where science is designed not just to generate knowledge, but to solve pressing public health challenges.

A strategic reorientation

A cornerstone of this transformation is the restructuring of the ICMR's institutional architecture. Recent reforms have expanded the mandates of several institutes, repositioning them as interdisciplinary hubs rather than narrowly defined entities. The evolution of institutes toward domains such as digital health and data science, child health, and women's health reflect the changing contours of India's disease burden and technological capabilities. Another important step is the creation of a network of regional National Institutes of Health Research across the country, from Dibrugarh in the Northeast to Jodhpur in the west. These institutes will conduct operational research with State and district health systems to ensure that relevant research is conducted and the output is used on the ground. These changes signal a strategic reorientation toward future-ready science,



Rajiv Bahl

Secretary to Govt. of India, Department of Health Research, Ministry of Health & Family Welfare and Director-General, Indian Council of Medical Research

where AI, genomics, and real-time data systems are integral to public health decision-making.

Equally important is the shift from compartmentalised functioning to a connected national research ecosystem. Institutes are now envisioned as resource centres that contribute to a shared national mission, ensuring that evidence generated in one setting informs action across the country. This systems-level thinking is critical in an era where health challenges, be it antimicrobial resistance, pandemics, or non-communicable diseases, are complex and interconnected. Similarly, the research funding ecosystem is also going through a redesign.

This integrated continuum marks a decisive shift from funding projects to enabling solutions. It is further reinforced by the **National Health Research Programme (NHRP)**, which has identified 15 priority areas ranging from antimicrobial resistance and tuberculosis to mental health, nutrition, and emergency care.

Technology, too, is playing a transformative role. The integration of AI into diagnostics, surveillance, and program implementation is helping bridge longstanding gaps between urban and rural healthcare. AI-enabled tools for tuberculosis and diabetic retinopathy screening are already supporting frontline health workers, while innovations such as AI-driven nutritional monitoring are improving program delivery at scale. The **F-Drone** initiative, which began with vaccine delivery and has expanded to transport critical medical supplies, exemplifies how cutting-edge technology can overcome geographic barriers and bring care closer to communities.

At the frontier of science, advances across the medtech gamut, from medical devices and diagnostics to next-generation vaccines and therapeutics, are enabling more targeted, patient-centric interventions. In parallel, the integration of

traditional knowledge systems through innovative, evidence-based models is gaining global recognition. Schemes such as **First in the World Challenge** complemented with platforms such as **MedTechMitra** and the **Medical Innovations Patient Mitra** initiative are accelerating the journey from research to commercialisation, ensuring that publicly funded science translates into affordable, accessible technologies for the people.

Towards universal access

The true measure of these reforms, however, lies in their impact on public health. Initiatives such as the **India Hypertension Control Initiative** have demonstrated how evidence-based strategies can transform chronic disease management at scale. Mission-mode programmes in emergency care, including mobile stroke units and rapid cardiac response systems, are redefining outcomes in life-threatening conditions. Expanded diagnostic networks and indigenous technologies are strengthening early detection and treatment across diseases ranging from cancer to infectious outbreaks. These efforts are closely aligned with the **National Health Policy 2017**, which emphasises preventive and promotive healthcare, universal access, and quality of care.

As we look to the future, the vision is clear. The ICMR will continue to serve as a catalyst, bringing together researchers, clinicians, policymakers, and the industry to build a resilient health ecosystem. The roadmap to 2047 will be shaped by advances in digital health, bio-manufacturing, and sustainable development, with a strong emphasis on capacity building and global collaboration. The transformation underway within the ICMR is an invitation to all stakeholders to participate in this journey and to ensure that science serves society in the most meaningful way.

The roadmap to 2047 will be shaped by advances in digital health, bio-manufacturing, and sustainable development, with a strong emphasis on capacity building and global collaboration

Homo erectus fossil yields secrets long thought to be beyond genetics

That modern humans carry DNA fragments from two other members of the genus *Homo* raised the possibility that they may also harbour genetic material from other extinct human species; but in the 4 years since the Denisovan genome was published, no new genome from any other extinct human relative has been uncovered – until now

Arjun Panchapakesan

For much of modern history, human evolution was thought to progress in linear fashion, with one species replacing another, one by one. The famous 'ape turning to human' cartoon illustrating a stooped primate gradually straightening into a modern man essentially captures this idea.

That picture began to change in the second half of the 19th century when scientists started to unearth human-like fossils. At the time, scientists used geological clues such as the depth at which a fossil was found and the rock layers surrounding it to determine the age of fossils. This method could establish which fossils were older or younger than others but did not reveal precise dates.

The timeline suggested an intriguing possibility, however. It appeared that these different kinds of humans had not followed one another in succession but had lived alongside each other. Radiometric dating techniques later confirmed this idea: multiple human species had coexisted on the earth at various points.

Series of surprises

In 2003, the Human Genome Project announced the first high-quality sequence of the human genome. Human beings finally possessed the secret to their own identity – the reason(s) they were unique. That sense of identity, however, proved to be short-lived.

In 2010, scientists published the genome of the Neanderthals, our closest extinct relatives. When they compared Neanderthal DNA with the genomes of living people, they uncovered a remarkable result. Most humans alive today carry around 1-2% Neanderthal DNA in their own genomes. In Africans, this percentage is slightly lower, but even their DNA is 0.3-0.5% Neanderthal.

The surprises did not end there. In 2012, scientists sequenced the genome of Denisovans, another human lineage, finding that some present-day populations, particularly in Oceania and parts of Southeast Asia, had 3-6% Denisovan DNA.

That we carry DNA fragments from two other members of the genus *Homo* raised the possibility that we may also harbour genetic material from other extinct human species. But in the 14 years since the Denisovan genome was published, no new genome from any other extinct human relative has been uncovered.

DNA after death

The difficulty lies in the nature of DNA itself. The moment an organism dies, its DNA begins to break down. Enzymes



Representative image of the bones of the *Homo erectus* fossil 'Turkana Boy', the most complete fossil of the species found to date. ERSABEN SCHOTZE (CC BY-SA)

released from dying cells cut DNA into small pieces. Microbes and fungi invade the remains and further degrade the genetic material. Water and oxygen promote chemical reactions that further damage DNA. Then there are temperature fluctuations, solar radiation, and UV damage. Over a few tens of millennia, the DNA is reduced to a few small pieces.

To recover DNA from these ancient humans, scientists need to be very lucky. The specimen must have been exposed to cold, dry, and stable environments such as permafrost, frozen sediments, or the inside of a deep cave. This is very rare – and why only a small fraction of fossils contain usable DNA.

One such species whose DNA sequence has evaded us for a long time is *Homo erectus*, the species estimated to have originated over 2 million years ago and thought to be one of the first human relatives to spread widely across Africa, Europe, and Asia.

Acid etching

A recent study published in *Nature* has provided the first molecular sequences from *Homo erectus* fossils from China. Rather than attempting to recover DNA directly, the researchers extracted proteins preserved within the enamel of six *Homo erectus* teeth dated to around

4,000,000 years ago. Because proteins are the products of the DNA, their sequences can reveal portions of the underlying genetic information, providing a rare, albeit narrow, view into the genome of an extinct human species.

The method used to recover the genetic information is notable. Usually, recovering ancient DNA or proteins requires scientists to grind up part of a fossil, so museum curators are reluctant to permit such analyses, especially when there is no guarantee that usable genetic material will be recovered.

To overcome this problem, the researchers used acid etching, a technique where a small area of the tooth enamel is briefly exposed to a dilute acid solution, which dissolves microscopic amounts of tooth enamel and releases the proteins trapped within it. The fossil is left largely intact. Since tooth enamel is a highly mineralised tissue, it can trap and protect proteins for a long time.

The researchers proceeded to successfully recover enamel proteins from five male and one female *Homo erectus* individuals. When they compared the protein sequences with those of modern humans, Neanderthals, and Denisovans, they made two interesting discoveries.

First, all six *Homo erectus* individuals had a protein variant that has never been

found in any other known species in the genus *Homo*. Second, they carried another variant present in Denisovans. This suggests populations related to these Chinese *Homo erectus* may have interbred with Denisovans in East Asia.

Shared story

However, because the results come from only the enamel proteins and not the entire genome, they are not conclusive. It was long known that the Denisovan genome contains DNA from a much older species of the human lineage. While this study points to *Homo erectus* as a possible candidate, it does not prove it.

Nevertheless, the study represents an advance far beyond the information it has generated, which by itself is tiny. This is the first effort to recover meaningful molecular data from *Homo erectus* fossils, long thought to be beyond the reach of genetics.

Though Neanderthals, Denisovans, and others have vanished from the earth, traces of their existence endure within our DNA. The more we look, the more we find that the story of humans is not only ours: it is theirs as well.

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THE GIST

Radiometric dating techniques have confirmed that multiple human species had coexisted on the earth at various points

While the Human Genome Project in 2003 provided humans with a sense of unique identity, later studies showed the presence of Neanderthal and Denisovan DNA in the modern human genome

A recent study on enamel from *Homo erectus* teeth has revealed a protein variant never found before in the genus as well as another variant present in Denisovans

It was known that the Denisovan genome contains DNA from a much older species of human lineage and the study, while not conclusive, points to *Homo erectus* as a possible candidate

Thank you

“The price of success is daily effort”



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